

VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING



Probability and Statistics

Assignment Project Report

Determining the Impact of 3D Printer Parameter Adjustments on Print Quality - Accuracy - Durability

Advisor(s):	Dr. Nguyen Tien Dung	
Student(s):	NGUYEN LE THANH HAO	1952242
	PHAM NHAT KHOI	2352623
	Nguyen Song Dat	1952646
	Nguyen Phuoc Thinh	1852765

HO CHI MINH CITY, AUGUST 2025



Member list & Workload

No.	Full name	Student ID	Contribution
1	NGUYEN LE THANH HAO	1952242	100%
2	PHAM NHAT KHOI	2352623	100%
3	Nguyen Song Dat	1952646	100%
4	Nguyen Phuoc Thinh	1852765	100%



Contents

1	Introduction	6
1.1	Project Overview	6
1.2	Research Objectives	6
1.3	Dataset Description	6
1.4	Report Structure	6
2	Data Cleaning and Preprocessing	7
2.1	Data Loading and Initial Inspection	7
2.2	Data Quality Assessment	7
2.3	Data Cleaning Process	7
2.4	Final Cleaned Dataset	7
3	Descriptive Statistics	8
3.1	Summary Statistics	8
3.2	Data Visualization	8
3.2.1	Histograms	8
3.2.2	Box Plots	8
3.2.3	Scatter Plots	8
3.3	Correlation Analysis	8
3.4	Group Comparisons	8
4	Methodology	9
4.1	Research Objectives	9
4.2	Statistical Methods	9
4.2.1	Multiple Linear Regression	9
4.2.2	Model Selection	9
4.2.3	Model Diagnostics	9
4.3	Implementation Approach	9
5	Results and Analysis	10
5.1	Model Development	10
5.1.1	Model 1: Roughness Analysis	10
5.1.2	Model 2: Tension Strength Analysis	10
5.1.3	Model 3: Elongation Analysis	10
5.2	Model Comparison	10
5.3	Model Diagnostics	10
5.4	Key Findings	10
6	Conclusion	11
6.1	Summary of Findings	11
6.2	Practical Implications	11
6.3	Limitations	11
6.4	Future Research	11



7	Data Source	12
7.1	Dataset Information	12
7.2	Variable Definitions	12
7.3	Data Availability	12
7.4	Acknowledgments	12

List of Figures

List of Tables

Listings



1 Introduction

1.1 Project Overview

1.2 Research Objectives

1.3 Dataset Description

1.4 Report Structure



2 Data Cleaning and Preprocessing

2.1 Data Loading and Initial Inspection

2.2 Data Quality Assessment

2.3 Data Cleaning Process

2.4 Final Cleaned Dataset



3 Descriptive Statistics

3.1 Summary Statistics

3.2 Data Visualization

3.2.1 Histograms

3.2.2 Box Plots

3.2.3 Scatter Plots

3.3 Correlation Analysis

3.4 Group Comparisons



4 Methodology

4.1 Research Objectives

4.2 Statistical Methods

4.2.1 Multiple Linear Regression

4.2.2 Model Selection

4.2.3 Model Diagnostics

4.3 Implementation Approach



5 Results and Analysis

5.1 Model Development

5.1.1 Model 1: Roughness Analysis

5.1.2 Model 2: Tension Strength Analysis

5.1.3 Model 3: Elongation Analysis

5.2 Model Comparison

5.3 Model Diagnostics

5.4 Key Findings



6 Conclusion

6.1 Summary of Findings

6.2 Practical Implications

6.3 Limitations

6.4 Future Research



7 Data Source

7.1 Dataset Information

7.2 Variable Definitions

7.3 Data Availability

7.4 Acknowledgments



References

- [1] Leo Breiman. Random forests. *Machine learning*, 45(1):5–32, 2001.
- [2] Thomas Cover and Peter Hart. Nearest neighbor pattern classification. *IEEE transactions on information theory*, 13(1):21–27, 1967.
- [3] Norman R Draper and Harry Smith. Applied regression analysis. *John Wiley & Sons*, 1998.
- [4] GeeksforGeeks. Random forest algorithm in machine learning, 2024. Accessed: 2024.
- [5] GeeksforGeeks. What are the advantages and disadvantages of random forest?, 2024. Accessed: 2024.
- [6] Selcuk University Research Group. 3d printer dataset, 2023. Kaggle Dataset for 3D Printer Quality Analysis.
- [7] Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The elements of statistical learning: data mining, inference, and prediction*. Springer Science & Business Media, 2009.
- [8] IBM. What is random forest?, 2024. Accessed: 2024.
- [9] Built In. Random forest: A complete guide for machine learning, 2024. Accessed: 2024.
- [10] Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani. *An introduction to statistical learning*. Springer, 2013.
- [11] M. Lichman. UCI machine learning repository, 2013.
- [12] Douglas C Montgomery, Elizabeth A Peck, and G Geoffrey Vining. *Introduction to linear regression analysis*. John Wiley & Sons, 2012.
- [13] J Ross Quinlan. *C4. 5: programs for machine learning*. Morgan kaufmann, 1993.
- [14] SkillCamper. Random forest: Why ensemble learning outperforms individual models, 2024. Accessed: 2024.